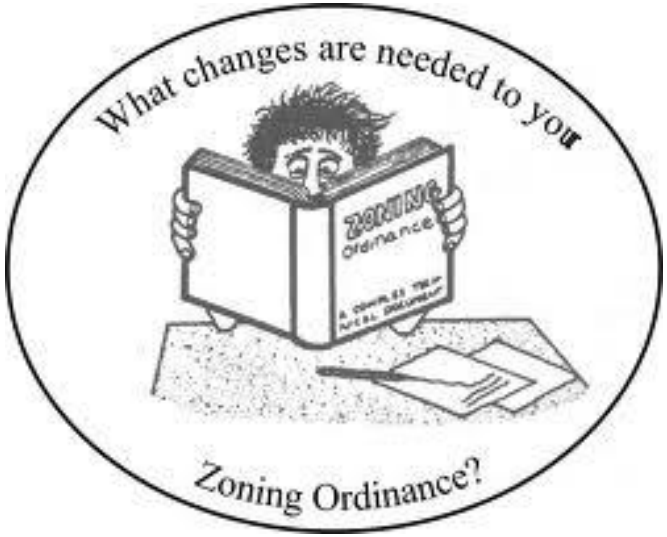
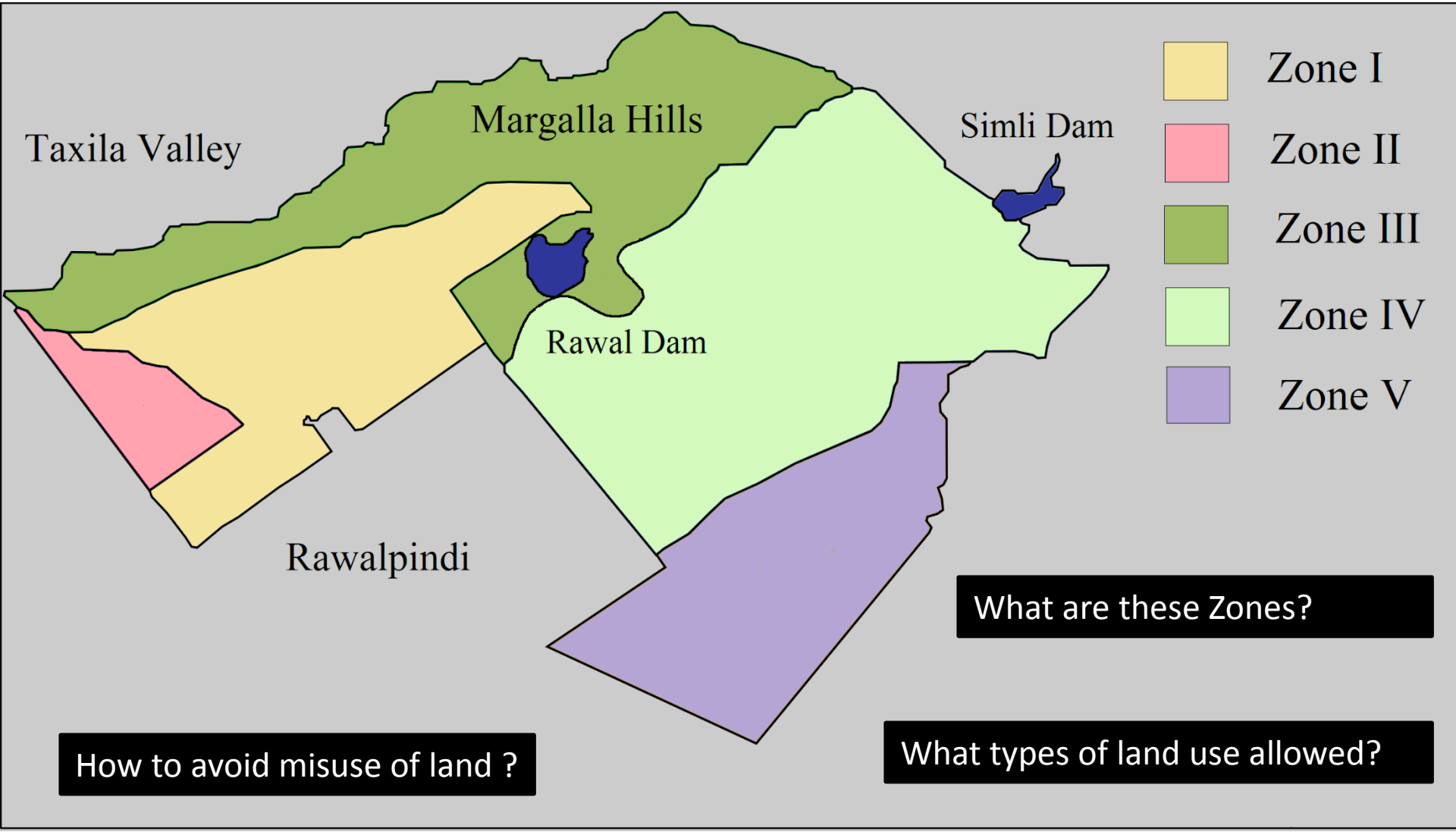
Zoning and Neighborhood Design



ARCH 305 Architecture and Town Planning

Contents

- Zoning
- Types of Zoning
- Zoning Standards
  - City/Town level
  - Housing Scheme
  - House



Land Use Policy provides the overall framework and general direction for development of a parcel

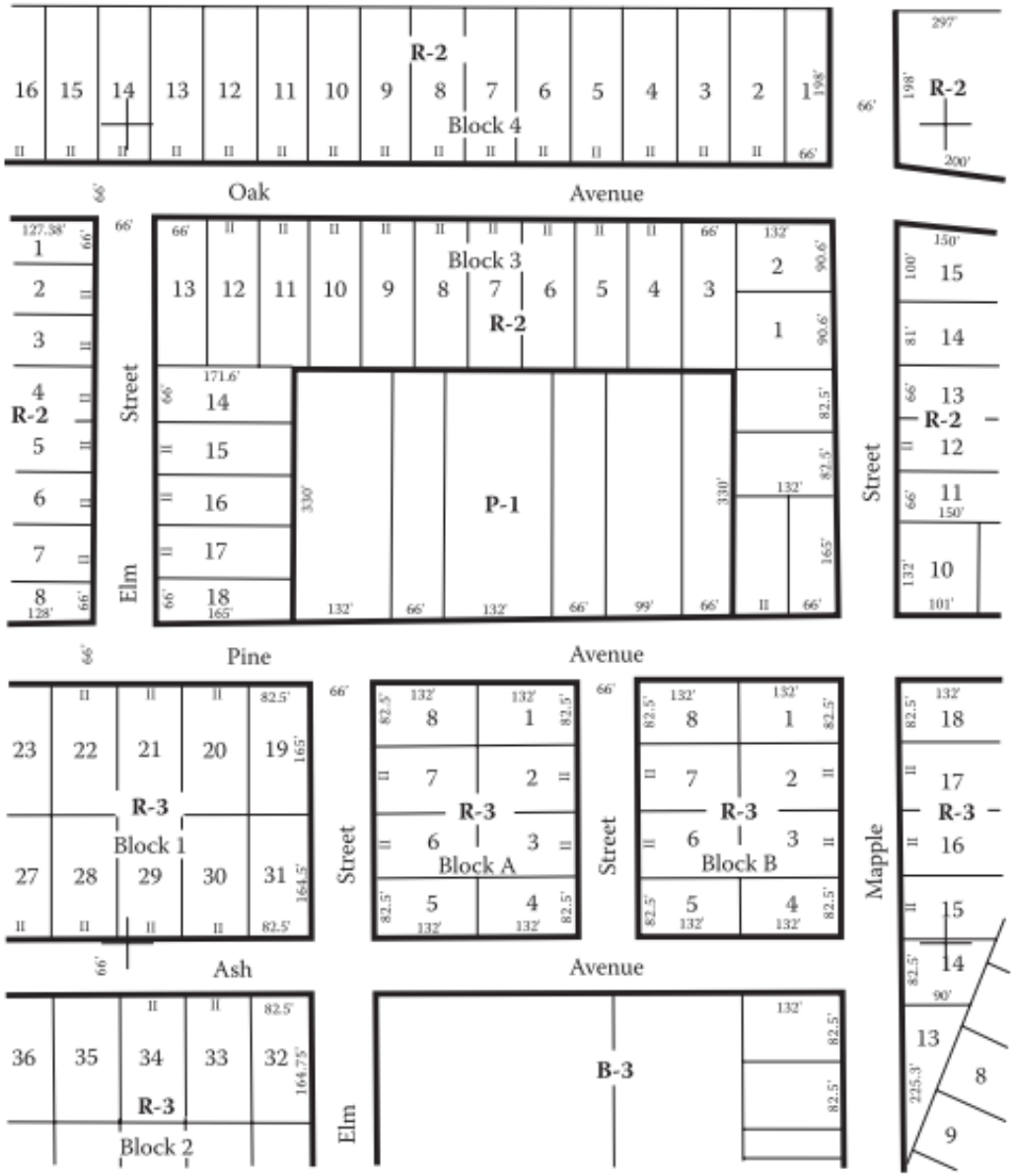
Zoning (1)

- Zoning is one of the most common methods of land use control used by local governments.
- Zoning implements Land Use Policy by detailing the specific development standards for individual parcels of land.
- Zoning has to be consistent with the Land Use Policy.

Zoning (2)

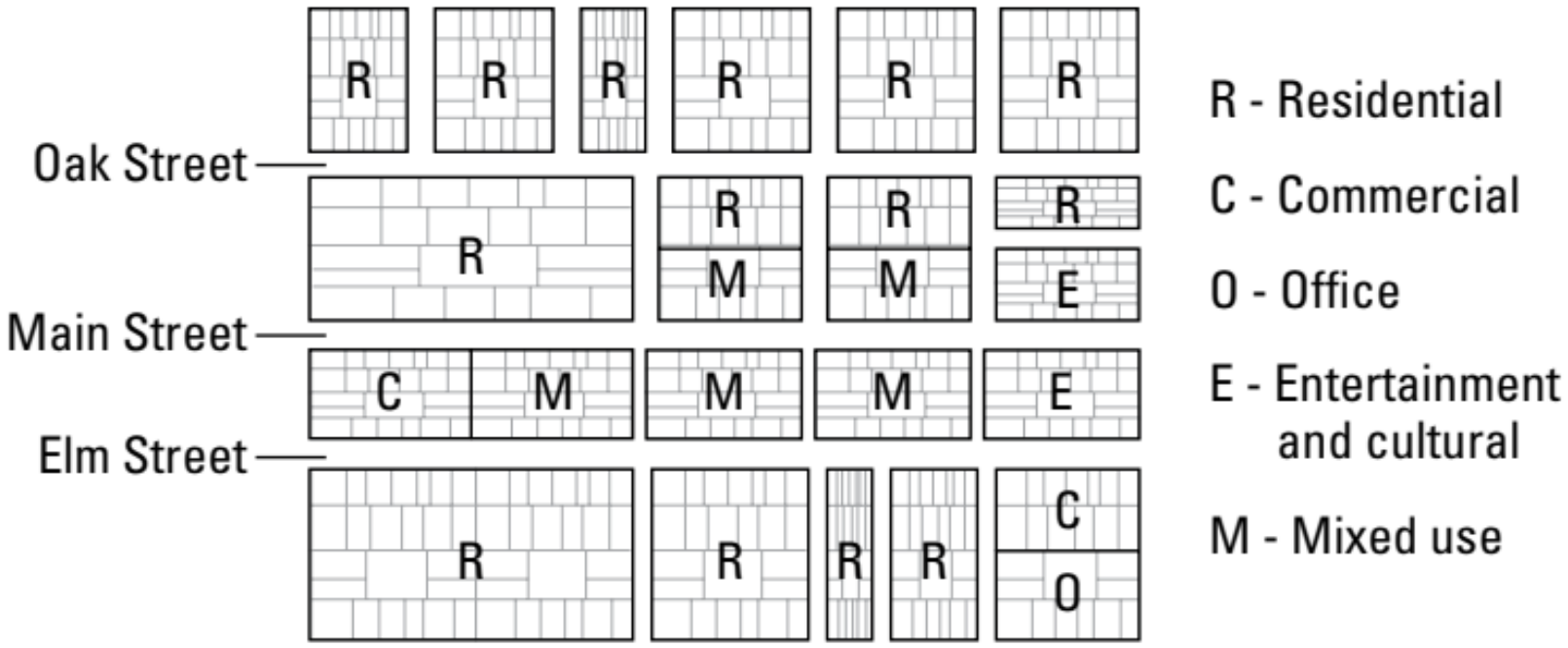
- Zoning is the division of a municipality, city or town into districts/zones for the purpose of regulating the use of private land.
- Zoning laws typically specify the areas in which residential, industrial, recreational or commercial activities may take place.
- For example, an R-2 residential zone might allow only single-family detached homes as opposed to duplexes or apartment complexes in R-3.

Example of a Typical Zoning Map



Legend

R-2	Single-family residential district	B-3	Integrated business district
R-3	Multi-family residential district	P-1	Public and semi-public district

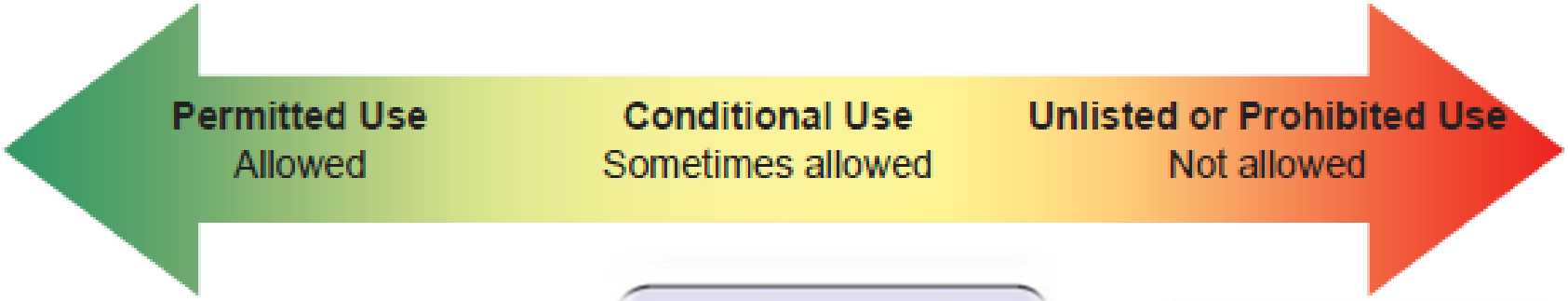

Oak Street
Main Street
Elm Street
R
C
M
E
O
R - Residential
C - Commercial
O - Office
E - Entertainment and cultural
M - Mixed use

Functions of Zoning (3)

- Determines the permitted uses on a parcel
- Spells out specific development standards regulating the use of individual parcels (i.e. setbacks, max heights, min lot sizes etc.)
- Has the force of law; provisions are enforced by the County
- Ensures that the exercise of an individual owner's property rights does not encroach upon the rights of another property

Zoning Ordinance (4)

Zoning ordinances **specify the permitted uses**, the building density, the required yards, offsets, right of way, off street parking and other prerequisites to obtaining permission.



Permitted Use - A use listed in the zoning ordinance that is allowed 'by right' at all locations in a zoning district.

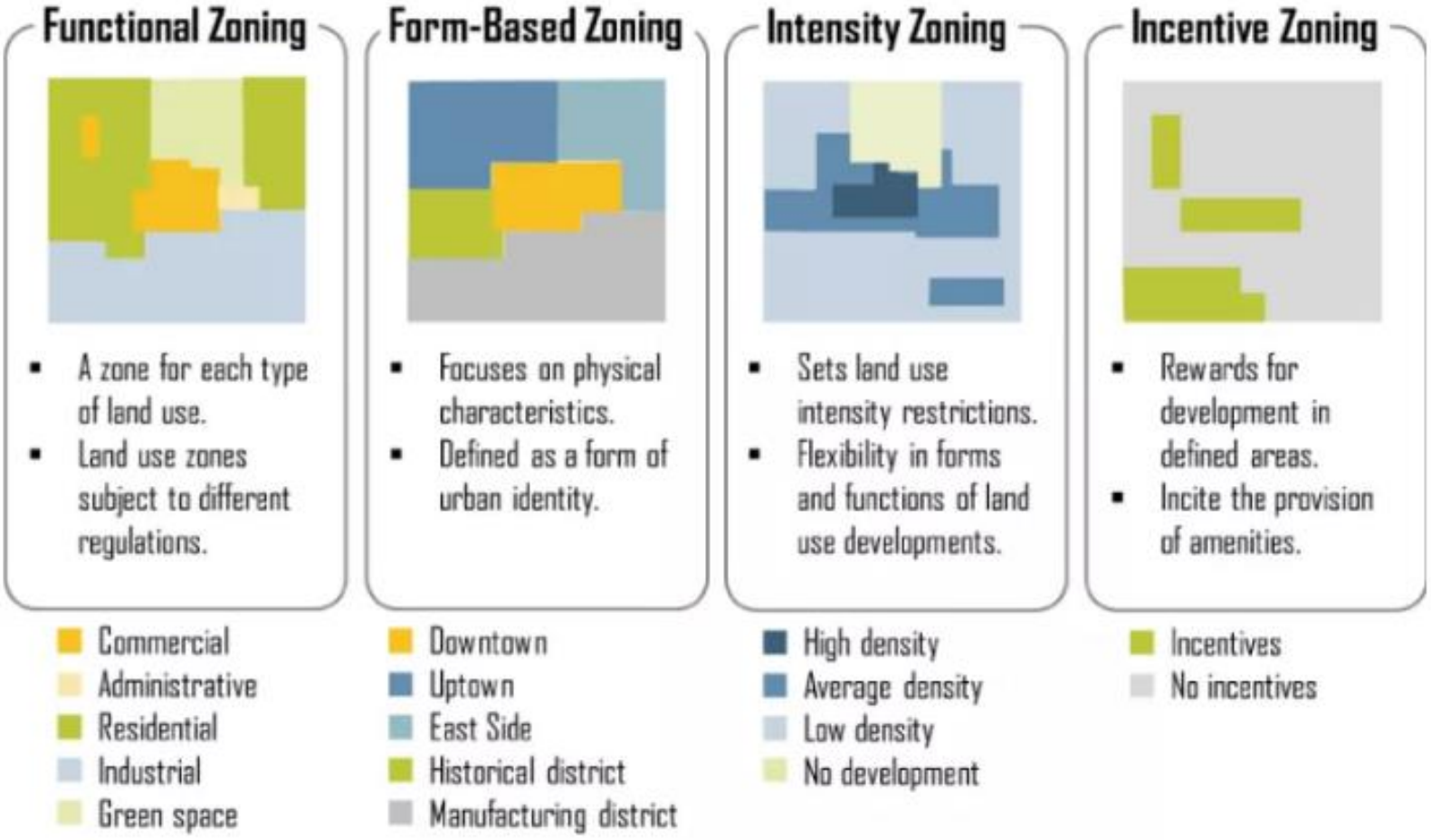
Conditional Use - A use listed in the zoning ordinance that may be allowed if found to be compatible with neighboring uses, limitations of the site, and the purposes of the ordinance. Conditions may also be attached upon approval.

Unlisted or Prohibited Use - A use that is not allowed in a district because it is not expressly listed or is specifically prohibited by the zoning ordinance.

Benefits of Zoning

- Effective way of plan implementation
- Prevents the misuse of land
- Prevents the encroachments
- Combine the similar zones & disparate zone to minimize the inter zonal effects(method of repulsion and attraction)
- Limits the height of buildings and densities of population in different zones
- Safeguard the environment

Zoning Types



Types of Land Use Zoning

Source: Adapted from Cambridge Systematics.

Classification of Zoning

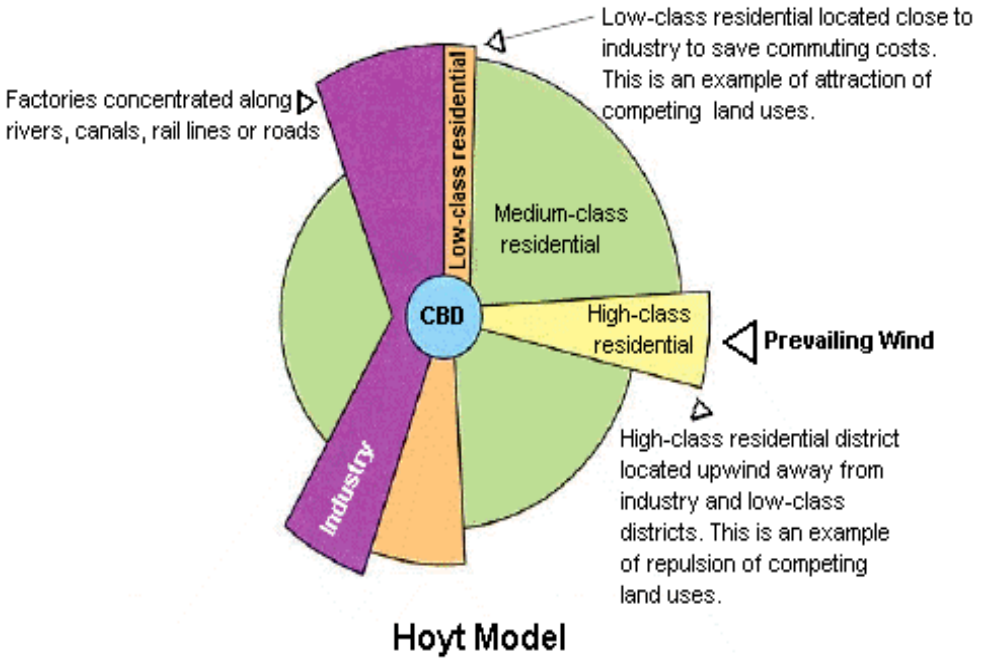
- Generally zoning is classified as;

  1. Functional Zoning (Land Use Districts)
  2. Height Zoning and
  3. Density Zoning (Intensity Zoning)

Use Zoning/Land Use Districts(1)

Use Zoning (Land Use Districts) divide the city/any area into different zones purposefully.

- Under this zoning a city/area can be divided into:
  - Residential Zone
  - Commercial Zone
  - Industrial Zone
  - Civic Zone
  - Institutional Zone
  - Recreational Zone



Hoyt Model

Use Zoning/Land Use Districts(2)

Residential Zone, a vital zone, contains buildings such as

- Detached single family houses
- Semi-detached houses
- Group housing
- Flats
- Sky-scrapers
- Parks





24-50 % of total land

- This zone should be away from business, industrial and have privacy.
- Should be separated from other zone via GREEN BELTS/Parks

Use Zoning/Land Use Districts(3)

- **Commercial zone:** this zone contains markets, ware-houses, godowns, business office, banks etc.



0.5-5 % of total land

- **This zone should be near to traffic nodes and have good transport connections**

Use Zoning/Land Use Districts(4)

- **Industrial Zone**: Usually contained light, medium and heavy industries and hence sub-divided accordingly.
- Can be planned near residential zone but then should have buffer zone in between

- Location of industries vital and direction of this zone should be away from town

2-20 % of total land

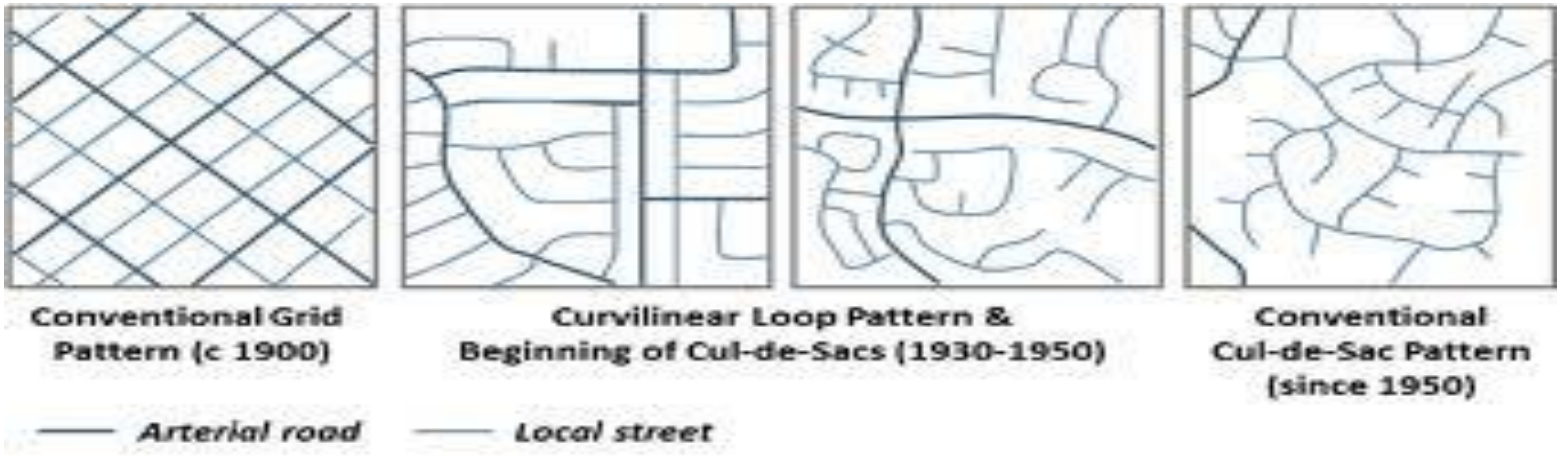
Use Zoning/Land Use Districts(5)

- **Institutional Zone**: Usually contains public buildings, schools, colleges, post office, banks, auditorium, courts etc.
- Can be planned near residential

2-21 % of total land

Use Zoning/Land Use Districts(5)

- **Arterial Circulation**: Usually contains networks of road such as primary road, secondary roads, highways, by-passes, streets etc



2-29 % of total land

Why developers don't allocate this proportion

Use Zoning/Land Use Districts(6)

- **Recreational Zone**: Usually contains Parks, open spaces, theme parks, gardens, etc
- It is vital as it maintains the aesthetic beauty of the area, safeguards the environment



0.5-7 % of total land

Use Zoning/Land Use Districts(7)

Proportion of Land use: City/town

TABLE 10.2 : OBSERVED RANGE OF LAND USE PROPORTIONS AT CITY/TOWN SCALE (%)

City/Town Population Size Class	Residential[1]	Industrial[1]	Commercial[2]	Institutional	Arterial Circulation/ Terminals	Recreational open spaces	Graveyards.	Vacant
All size Classes	24-50	2–20	0.5–5	2–21	2–29	0.5–7	0.5–6	3–45
500,000 +	24-32	2–15	1–2	3–8	13–20	2–5	0.5–3.5	9–45
100,000 – 499,000	26–48	3–8	0.5–2	2–10	12–29	1–7	0.5–4	3–17
50,000 – 99,000	27–43	2-20	1–5	3–11	3–27	1–6	0.5–6	8–26
25,000 – 49,000	26–50	3-11	0.5–3	2–21	2–18	0.5–2	1–4	7–31

1. Includes local networks and services.
2. Includes local networks and residential use not on ground floor.

Source: H&PP Dept., Lahore.

Use Zoning/Land Use Districts(8)

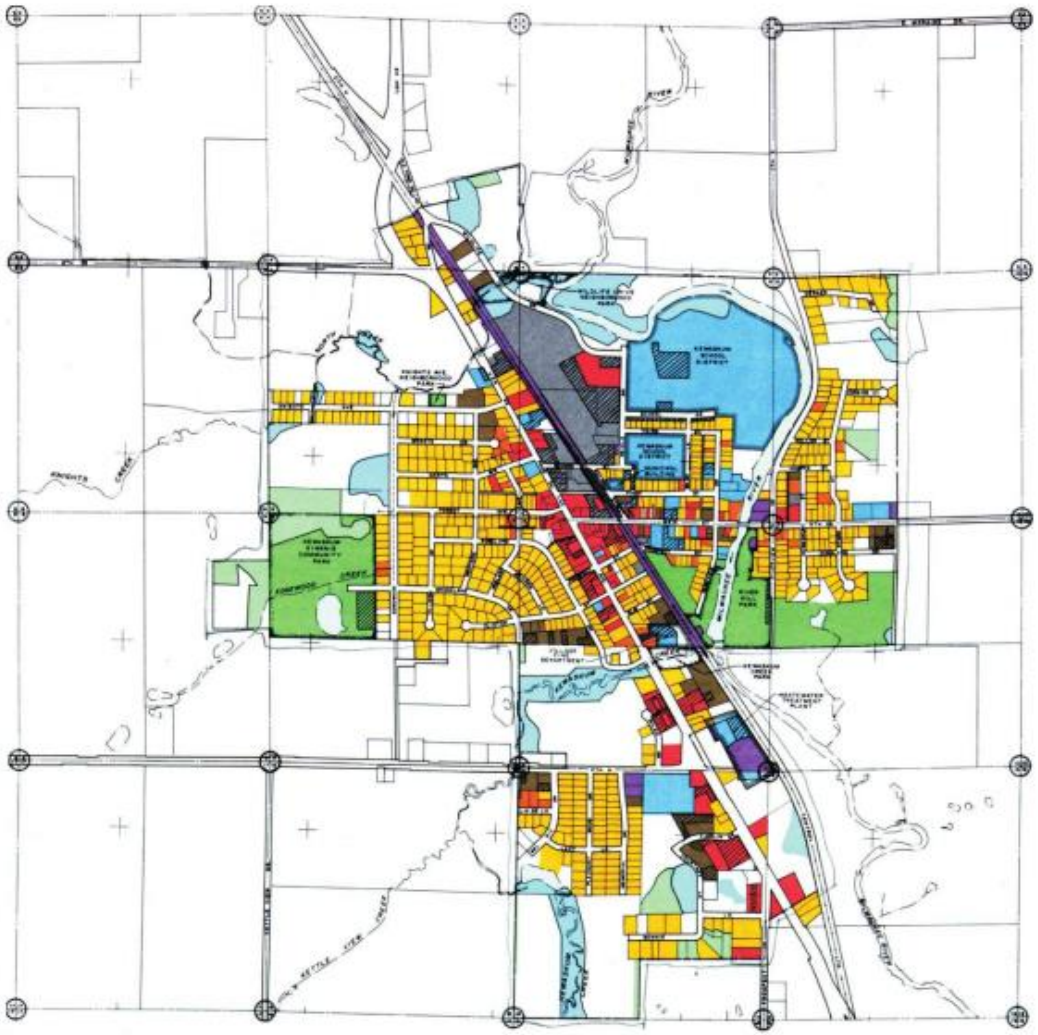
Proportion of Land use: H/Scheme

TABLE 10.1 : LAND USE DISTRIBUTION IN AREA DEVELOPMENT SCHEMES

Use	Percentage
i) Residential	45-52
ii) Commercial	2-3
iii) Education, Health & other Community Facilities.	7.5-10
iv) Roads/Streets	25-30
v) Open Spaces	5-7.5
vi) Others, including graveyards equivalent reserve*	2-5

*Reference 6.5.8.

Example of an Existing Land Use Map



Legend

- Single-family residential
- Two-family residential
- Multi-family residential
- Commercial
- Parking
- Industrial
- Transportation and utilities
- Governmental and institutional
- Recreational
- Wetlands
- Woodlands
- Agricultural and other open lands
- Surface water

Graphic Scale
0 400 800 1600 Feet

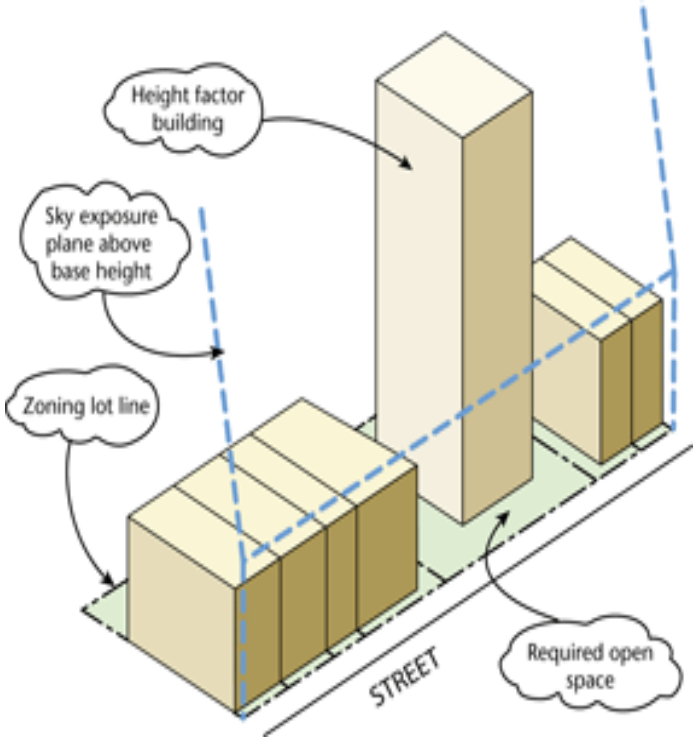
Use Zoning/Land Use Districts(9)

Standards for a residential building (CDA)

TYPE OF DEVELOPMENT	PLOT SIZE (Sq.Yd.)	FRONTAGE (Feet)	Max. GROUND COVERAGE (%)	Max. FLOOR AREA RATIO (FAR)	Max. STOREYS & HEIGHT		Min. SET-BACKS (Feet)				Max. HOUSING UNITS	Maxi CAR PORCH AREA (sft.)	Max. AREA of Mumty(s) (sft)	BASEMENT	Max. PLINTH LEVEL
					No.	Height	Front	Side-1	Side-2	Rear					
TERRACED (ATTACHED) & SEMI-DETACHED DWELLING HOUSES															
TERRACED / ATTACHED Type 'A'	Up to 150	20-29	Construction may cover a block, leaving compulsory setbacks		2	30'	5'	--	--	5'	One	100***	200 sft.	Allowed leaving the setbacks	3'-6"
	151 to 200	25-30			2	30'	6'	--	--	6'	One	180***			
	201 to 320	30-39			2	30'	10'	--	--	8'	One	180 **	350 sft.		
	300 to 450	40-49			2	30'	10'	--	--	10'	One	200 **			
SEMI-DETACHED Type 'B'	300 to 625	40-49			2	30'	10'	--	5' *	10'	One	200 **	350 sft.		
DETACHED DWELLING HOUSES															
DETACHED Type 'C'	400 to 1000	50 to 59	Construction may cover a block, leaving compulsory setbacks		2	30'	15'	5'	5'	10'	Two	Single (200 sft) or Double (400 sft) per unit is allowed. **	Mumty of 400 sft.	Allowed leaving the setbacks	5'-0"
	530 to 1335	60 to 69			2	30'	15'	10'	5' *	10'	Two				
DETACHED Type 'D'	700 to 1670	70 to 79			2	30'	20'	10'	10'	10'	Two				
	885 to 2670	80 to 89			2	30'	25'	10'	10'	10'	Two				
	800 to 2900	90 to 99			2	30'	30'	10'	10'	10'	Two				
	1770 to 2720	100 +			2	30'	40'	15'	15'	10'	Two				
FLATS / APARTMENT BUILDINGS															
FLATS / APARTMENTS	Type 'A' Apartments (Up to Approx. 2.0 Acres)		35 %	2.8	8	92'	80'	30'	30'	40'	- do -	N.A	3000 sft.	- do -	5'-0"
	Type 'B' Apartments (More than 2.0 Acres)		30 %	3.6	12	135'	80'	35'	35'	40'	- do -	N.A	4000 sft.	- do -	5'-0"

Height Zoning (1)

- height and density regulations for buildings are significant in zoning.
- Height and volume of the building is controlled by Height Zoning.
- The purpose is to ensure adequate sunshine, air, breeze etc to small units



Methods

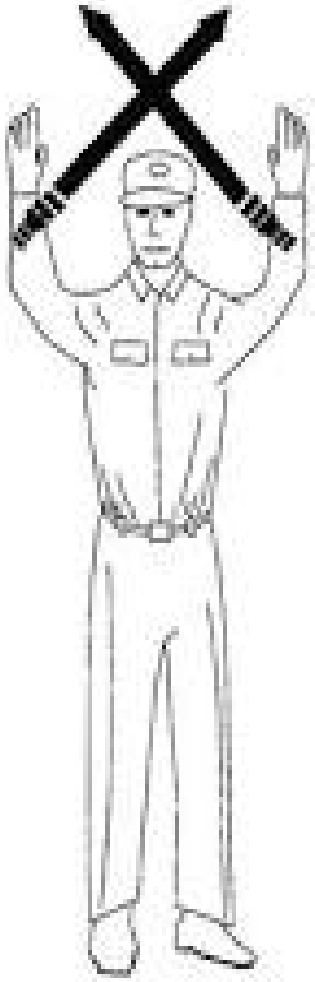
Density and Bulk Control

45 and 63.5 degree Air Plane Rule

45 and 63.5 degree Air Plane Rule

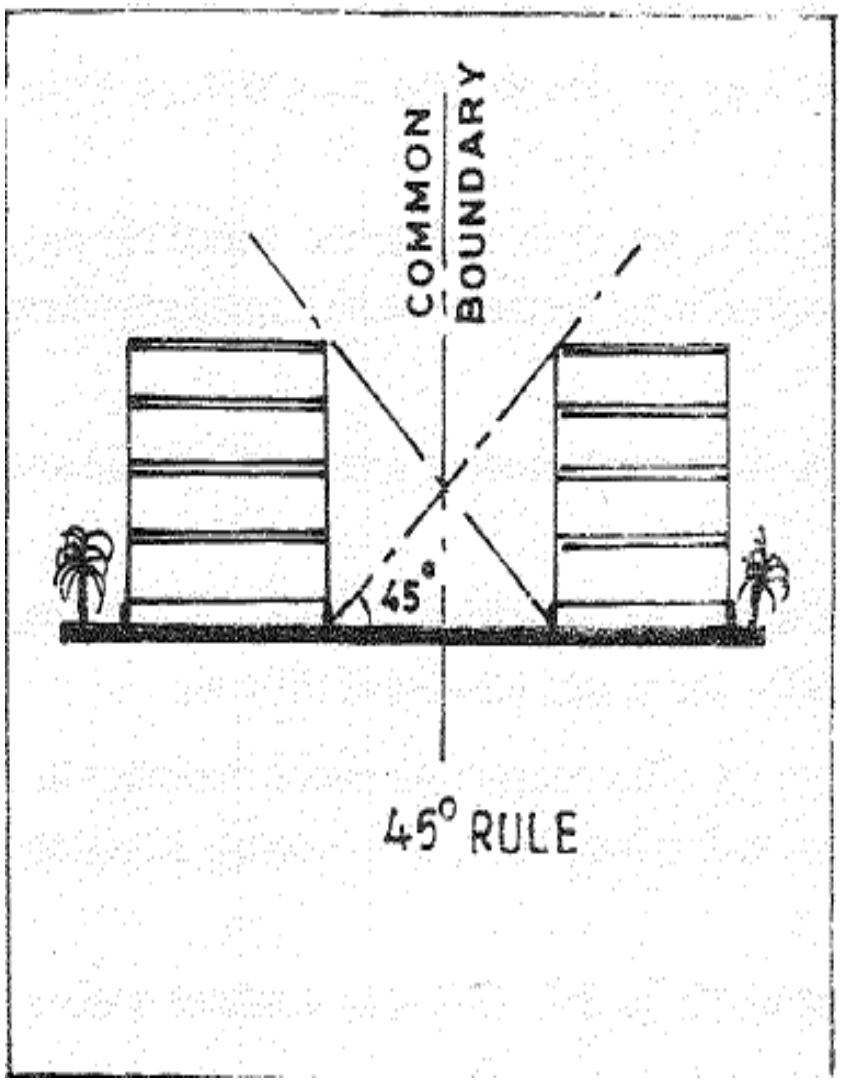
Tool to control height

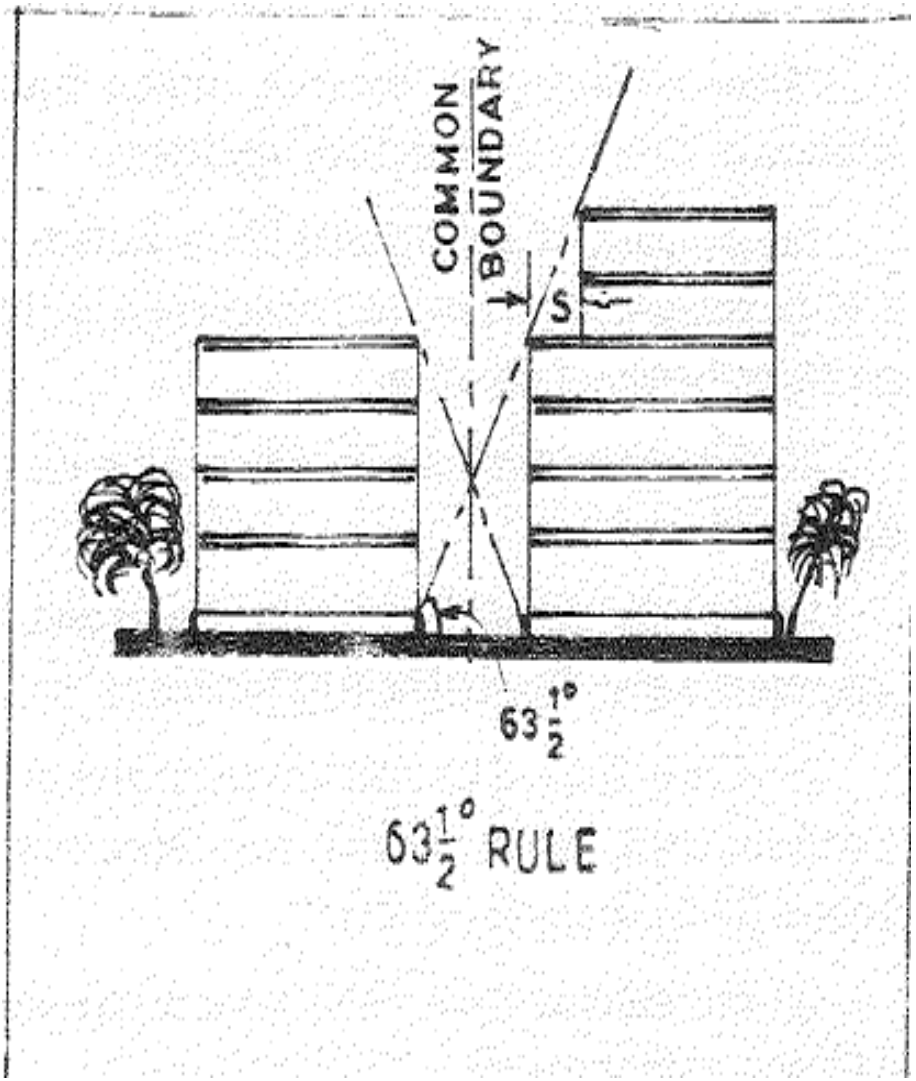
- In this method height of building is regulated according to the width of the abutting road
- The rule is that no part of tall building should cut the plane drawn from the boundary of the plot at an angle of 45 or 63.5 degree to the horizontal.
- In case of 45 degree height : road width is 1:1 and 2:1 in case of 63.5degree



45 and 63.5 degree Air Plane Rule

Tool to control height





Density and Bulk Control

ZONING BULK & HEIGHT REGULATIONS
FOR RESIDENTIAL PLOTS

Zoning District	Type of development	Plot Size		BUILDINGS				Max. permissible covered area of stair tower	MIN. SET BACK				Remarks
		Area in Sq. Yd.	Frontage in feet	Floor area Ratio	Max. built up area on ground Floor	Max. No. of storeys	Max. height		Front	Side-1	Side-2	Rear	
All private residential plots in Islamabad allowed by the Authority except those in the Diplomatic Enclave.	Terraced dwelling houses Type 'A'	upto 150 151 - 200	20 - 29 25 - 30	1.2 1.2	60% 60%	2 2	30' 30'	150 Sft.	5' 6'	- -	- -	7' 8'	No allowance for stairs shall be given
	Terraced dwelling houses Type 'B'	201 - 625	30 - 49	1.05	60%	2	30'	200 Sft	10'	-	-	10'	
	Detached dwelling houses Type 'C'	400 - 1000 530 - 1335 700 - 1670	50 - 59 60 - 69 70 - 79	0.90 0.90 0.90	50% 50% 50%	2 2 2	30' 30' 30'	250 Sft	10' 10' 20'	5' 10' 10	10' 10' 10'	10' 10' 10'	
	Detached dwelling houses Type 'D'	885 - 2670 800 - 2900 1770 - 2720	80 - 89 90 - 99 100 & above	0.85 0.85 0.85	50% 50% 50%	2 2 2	30' 30' 30'	250 Sft	20' 20' 30'	10' 10 15	10' 10' 15'	15' 15' 15	

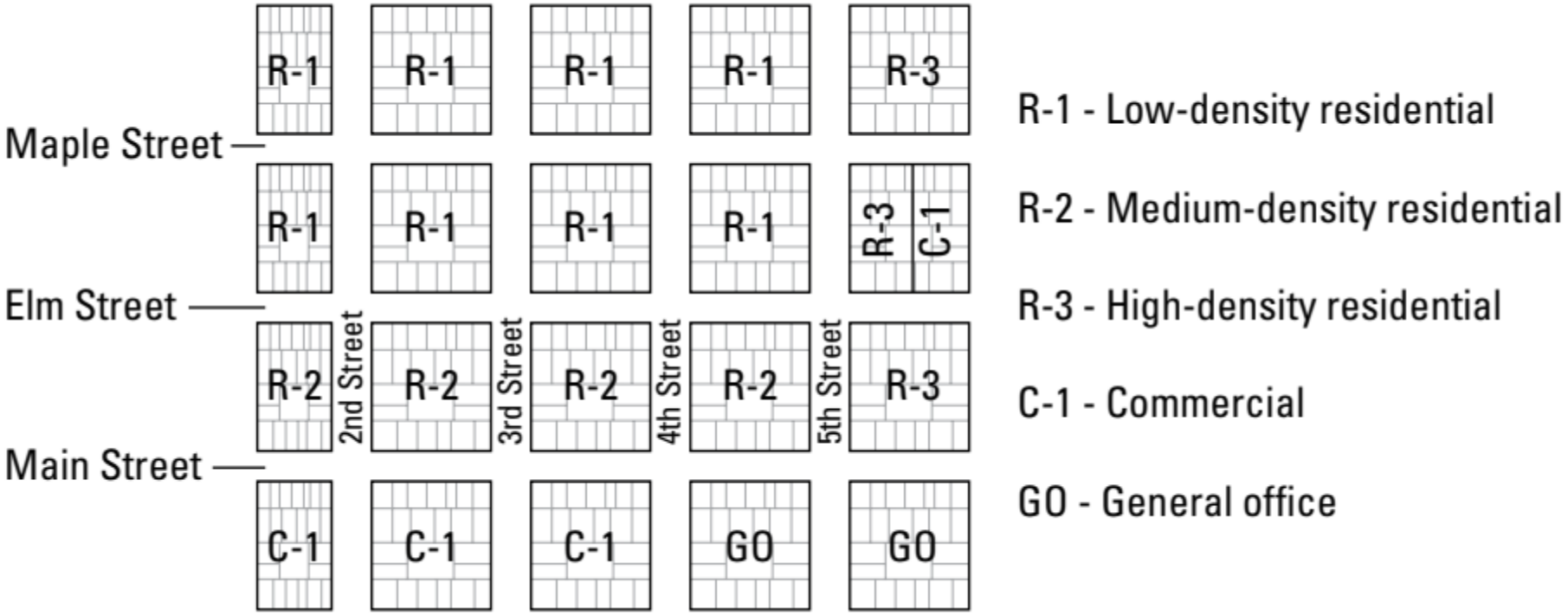
Note: Entire area of the structure measured externally shall be counted into covered area.

Density and Bulk Control

MINIMUM SPACE STANDARDS/SIZES

ANNEXTURE - '1'

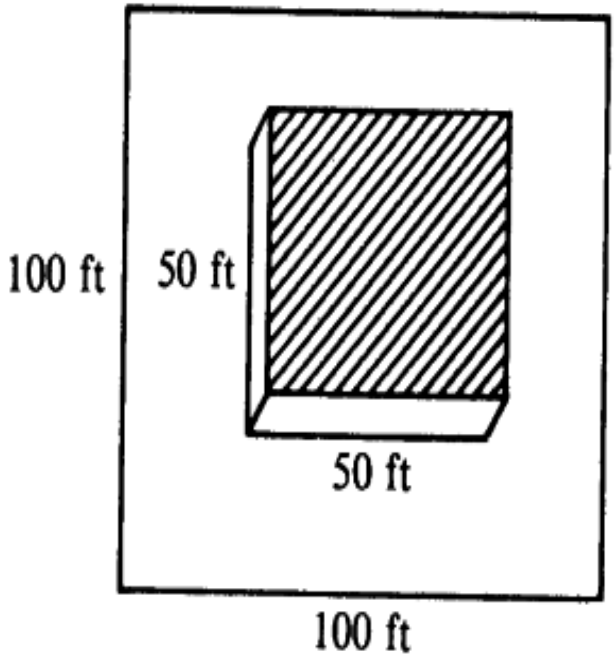
	Space	Minimum Area	Relaxable with Penalty	Minimum height	Relaxable with penalty	Minimum width allowed	Relaxable with penalty
1.	Living Unit	600 Sft. Excluding stairs	10%	9' - 0"	8' - 6"	-	-
2.	Habitable Rooms, i.e. Bed, Drawing, Lounge and servant room etc.	100 Sft.	10%	9' - 0"	8' - 6"	8' - 6"	8' - 0"
3.	Kitchen	45 Sft.	10%	8' - 6"	8' - 0"	5' - 0"	4' - 9"
4.	Baths / Toilet	24 Sft..	Upto 10%	7' - 0"	6' - 6"	3' - 0"	2' - 9"
5.	W. C.	12 Sft..	Upto 10%	7' - 0"	6' - 6"	3' - 0"	2' - 9"
6.	W. C. & W. B. both	15 Sft.	Upto 10%	7' - 0"	6' - 6"	3' - 0"	2' - 9"
7.	Shower only	12 Sft.	Upto 10%	7' - 0"	6' - 6"	3' - 0"	2' - 9"
8.	Powder Room/toilet under the stair /landing (W. C & W. B only properly ventilated)	15 Sft.	Upto 10%	6' - 6"	6' - 0"	3' - 0"	2' - 9"
9.	Garrage and Porches			7' - 6"	7' - 0"	9' - 0"	8' - 6"
10	Basement	If used for habitable purposes the rules for respective space uses will be applied.					
			Minimum height		Relaxable without fine upto		Relaxable with fine upto
11.	Doors		7' - 0"		6' - 6"		6' - 0"
12.	Head room under stairs if access/ passage is provided		6' - 6'		6' - 3"		6' - 0"
13	Width of stair flight		3' - 3"		3' - 0"		2' - 9"


R-1
R-1
R-1
R-1
R-3
Maple Street
R-1
R-1
R-1
R-1
R-3
C-1
Elm Street
R-2
2nd Street
R-2
3rd Street
R-2
4th Street
R-2
5th Street
R-3
Main Street
C-1
C-1
C-1
GO
GO
R-1 - Low-density residential
R-2 - Medium-density residential
R-3 - High-density residential
C-1 - Commercial
GO - General office

Floor Area Ratio

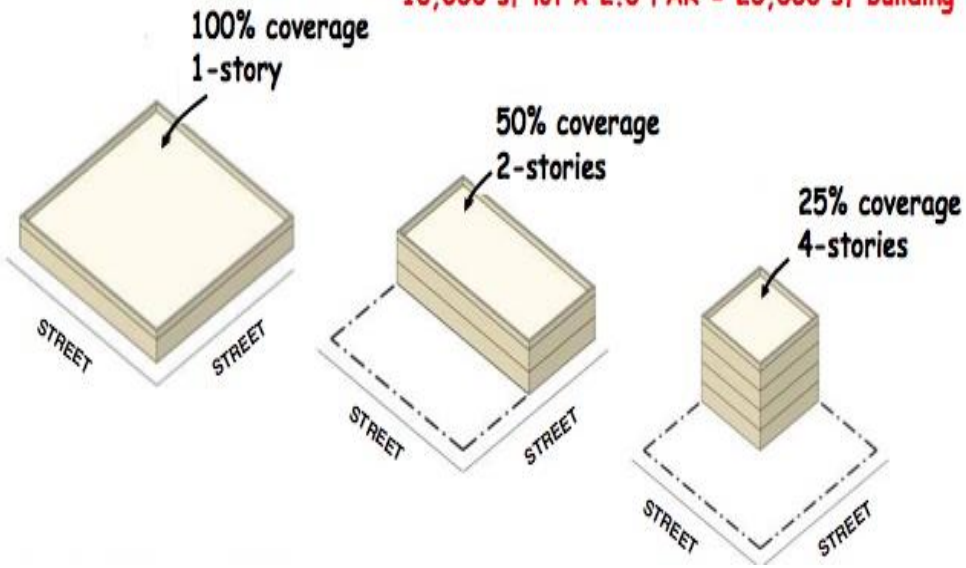
Tool to control height and intensity

FAR is the ratio between building's total floor area **to the size of the piece of land upon which it is built.**



Every zoning district has a **floor area ratio** (FAR). Multiplying the FAR by the **lot size** will give you the permitted **floor area** (size) of a building.

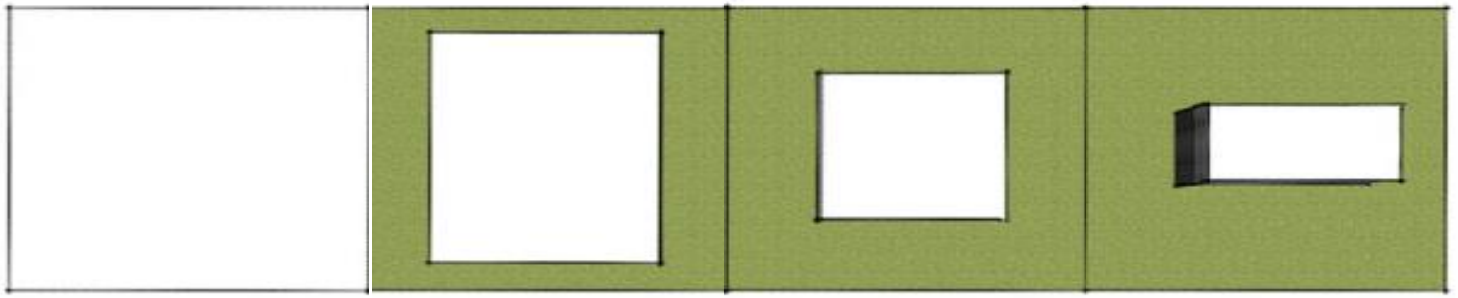
10,000 sf lot x 1.0 FAR = 10,000 sf building
10,000 sf lot x 2.0 FAR = 20,000 sf building



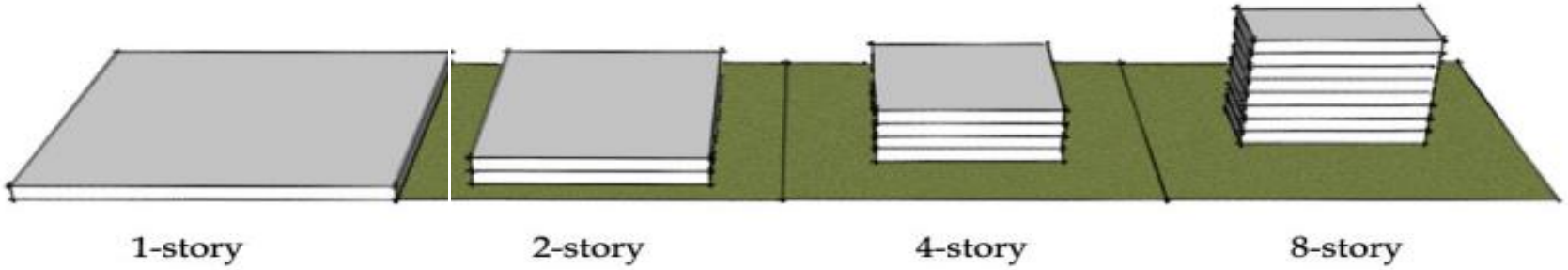
Floor Area Ratio

Tool to control height and intensity

Site Coverage



Building Height



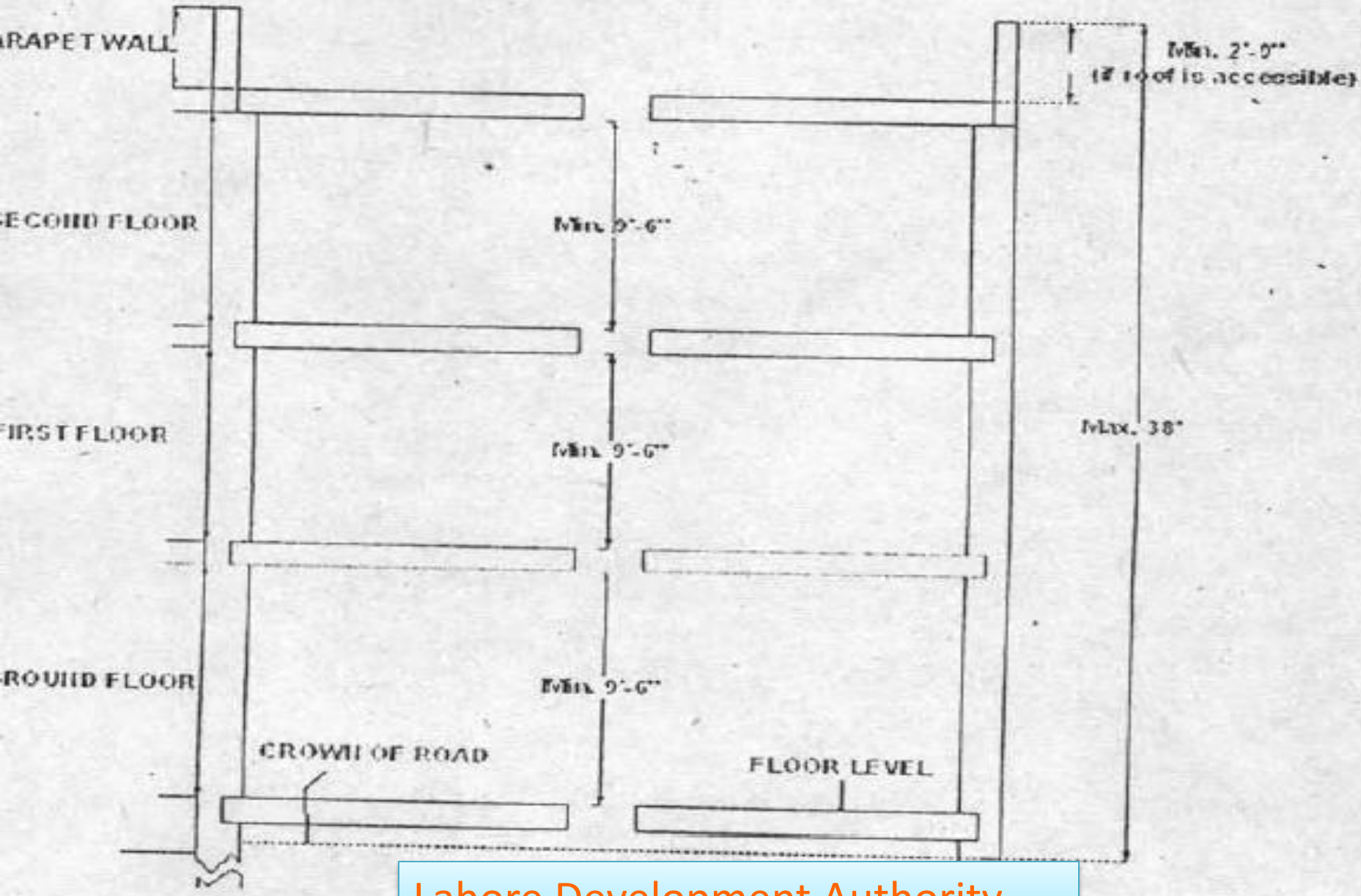
Lahore Development Authority

Mandatory Open Spaces:

a. For a building abutting on roads having up to 25 ft (7.62 m) right of way the building line shall be as established by the buildings which are in existence but as far as side and rear spaces are concerned, the following table shall be applied.

b. For the buildings abutting on roads having more than 25 ft (7.62 m) right of way the mandatory open spaces shall be as follows:

Plot Size/ Zone	Building Line	Rear Space	Side Space
Less than 5 Marlas	5 ft (1.52 m)	Not required	Not required
5 Marlas & above but less than 10 Marlas	5 ft (1.52 m)	5 ft (1.52 m)	Not required
10 Marlas to 30 Marlas	10 ft (3.05 m)	7 ft (2.13 m)	5 ft (1.52 m) (on one side)
Above 30 Marlas but less than 2 Kanals	10 ft (3.05 m)	7 ft (2.13 m)	5 ft (1.52 m) (on both sides)
2 kanals & above	20 ft (6.1 m)	10 ft (3.05 m)	10 ft (3.05 m) (on both sides)
Apartment building* on four kanals and above	30 ft (9.15 m)	13 ft (3.96 m)	13 ft (3.96 m) (on both sides)



Lahore Development Authority

Fig-2.1 Height of Residential Building

Lahore Development Authority

2.1.2 Building Height

a. The height of any building other than Apartment Buildings measured from the crown of the road to the top of the parapet wall (exclusive of chimney stacks, lift heads and water tower) shall not exceed 38 ft (11.58 m).

b. In case of Apartment Building the maximum building height allowed on residential plots measured from the crown of the road to the top of the parapet wall (exclusive of chimney stacks, lift heads and water tower) shall not exceed 45 ft (13.72m).

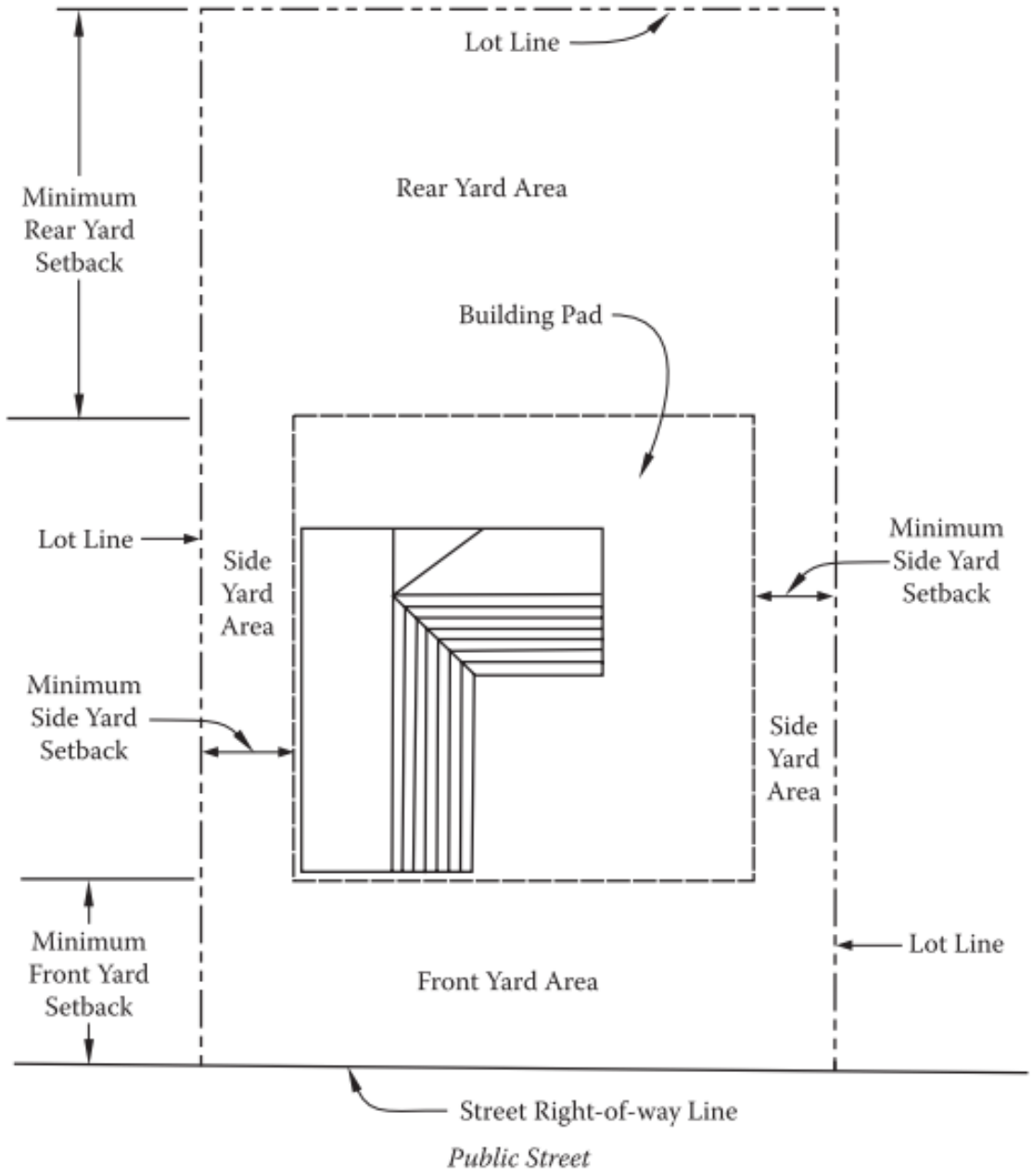
c. The total number of storey permissible in an apartment building on residential plots, excluding basements, shall not be more than four. Each storey shall have a minimum building height of 9-ft 6-inches (2.9m), other than the basement.

2.1.3 Ground Coverage and Floor Area Ratio (FAR)

For buildings abutting on roads having more than 25 ft (7.62m) right of way, the FAR and ground coverage shall be as applicable in approved schemes. Whereas FAR and ground coverage on roads having right of way up to 25 ft (7.62m) shall be as follows:

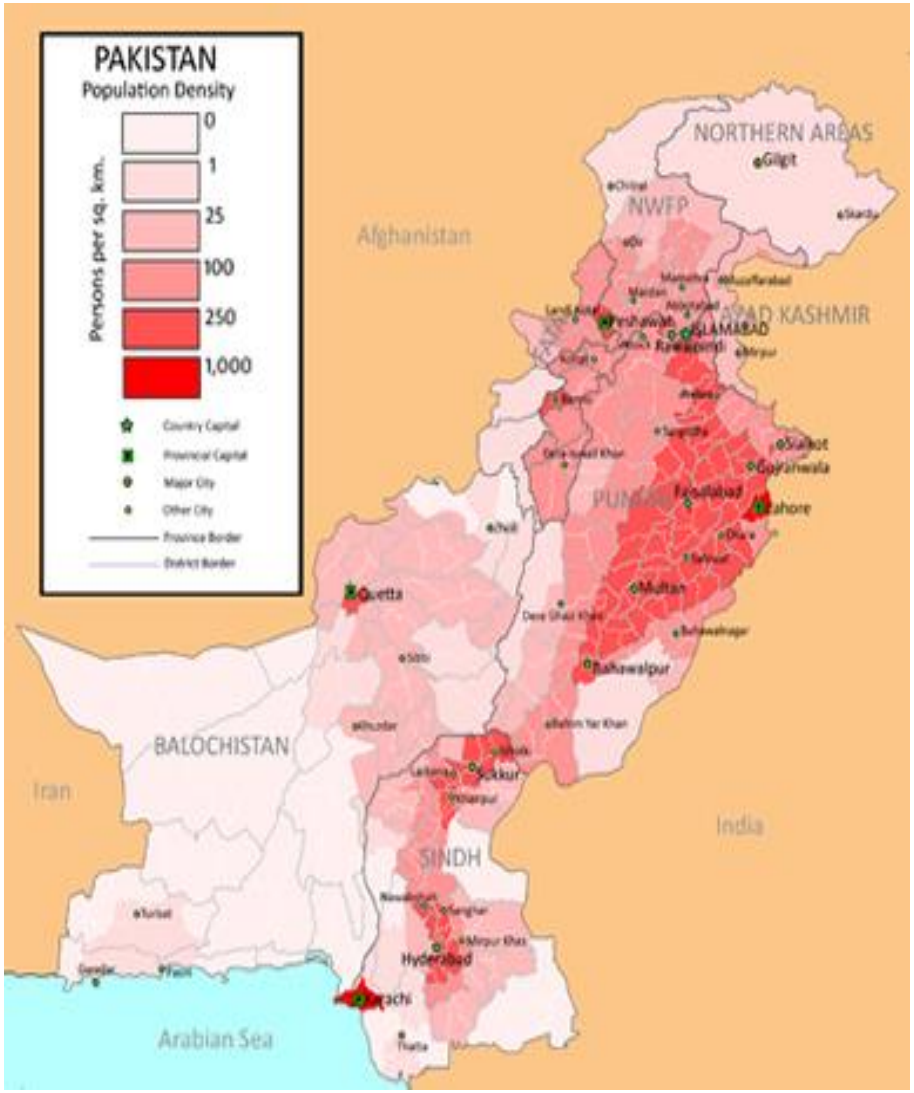
Plot Size	Maximum Ground Coverage	Maximum FAR
Less than 5 Marlas	85%	1:2
5 Marlas & above but less than 10 Marlas	80%	1:1.6
10 Marlas & above but less than 1 kanal	70%	1:1.5
1 kanal & above but less than 2 kanals	65%	1:1.4
2 kanals & above	60%	1:1.3
Apartment building on residential plot	55%	1:2.5

Location of Required Setbacks and Yards on a Typical Interior Lot



Density Zoning (1)

- The population per unit area or acres is defined as density of population.
- This density may be
  - gross density
  - net density



Density Zoning (2)

- The **net density** is the average density per unit of the housing area.

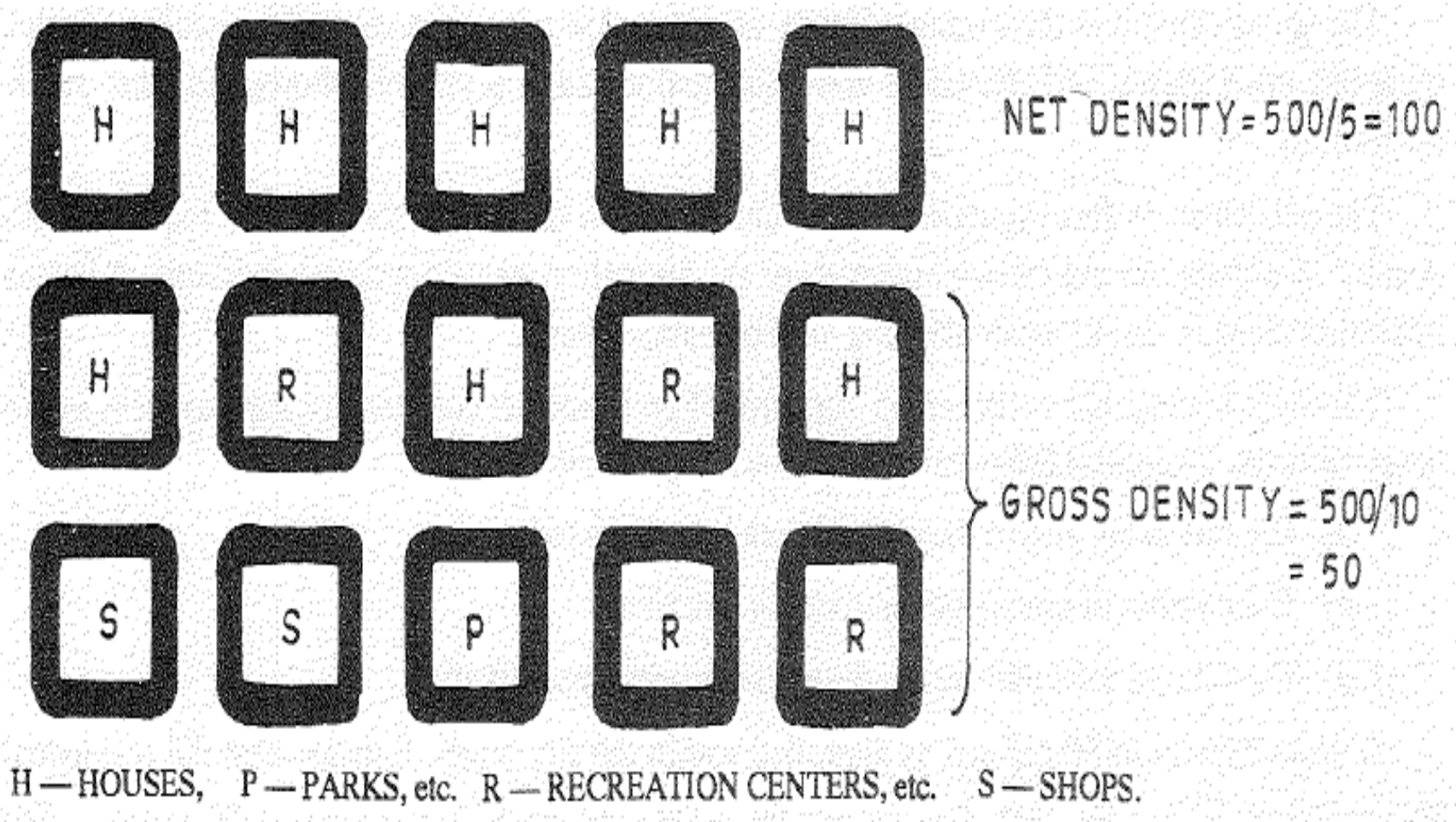
- **The gross density** is average density of population per unit area of the **residential zone**, including open spaces, and area under non-residential use such as schools, shops, and other public institutions

Density Zoning (3)

Example

- Consider a family consisting of 5 persons living in a plot,
  - the population in 100 plots will be 5 x 100 = 500.
  - If 20 plots are arranged per acre, then
  - land required will be ***100/20* = 5 acres.**
- **Hence the net density of population per acre will be 500/5 = 100**.
- If an area of 5 acres is reserved for playgrounds, recreation, then the gross density of population per acre will be 500/10 = 50.

Density Zoning (4)




THANK
YOU!